# ParaBank - Fund Transfer Module

## Test Cases by Scenario

## TS_FT_001: Verify fund transfer with valid accounts and sufficient balance

| Field | Details |
|---|---|
| **Test Case ID** | TC_001 |
| **Test Case Title** | Fund transfer with valid accounts and sufficient balance |
| **Preconditions** | User is logged in to ParaBank (russell / test1234)<br>User has at least two active accounts<br>Source account balance is $415.50 |
| **Test Steps** | 1. Navigate to Transfers > Transfer Funds<br>2. Select account 15123 (checking) from 'From Account' dropdown<br>3. Select account 16123 (savings) from 'To Account' dropdown<br>4. Enter amount: $100.00<br>5. Click the 'Transfer' button |
| **Test Data** | Username: russell \| Password: test1234<br>From Account: 15123 (Balance: $415.50)<br>Amount: $100.00 |
| **Expected Result** | Transfer confirmation page is displayed<br>Message shown: 'Transfer Complete!'<br>From account 15123 balance decreases by $100.00<br>To account 16123 balance increases by $100.00 |
| **Actual Result** | Transfer confirmation page displayed immediately<br>Message 'Transfer Complete!' visible with amount and both account numbers<br>Source account balance updated to $315.50<br>Destination account balance receives $100.00 |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Critical |

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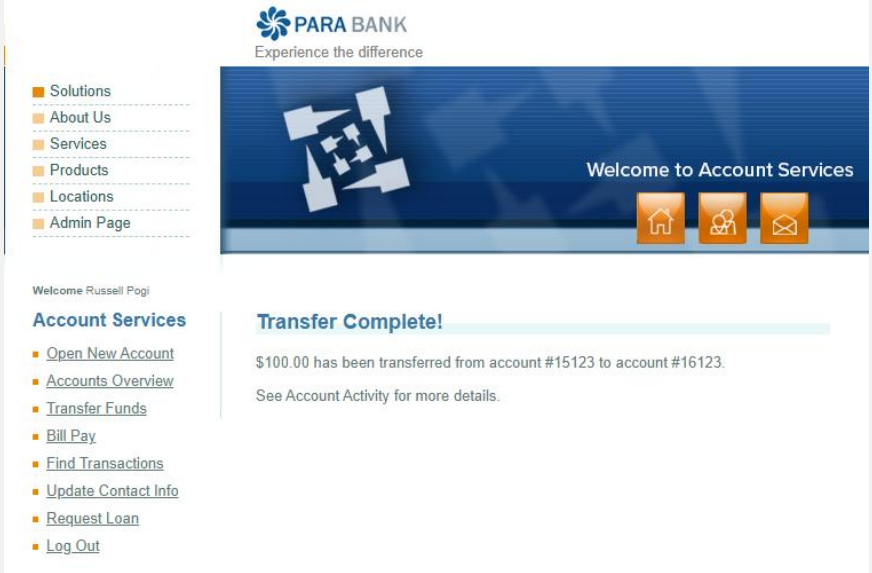
|---|---|
| **Remarks / Comments** |  <br>Transfer details on confirmation page are accurate. Balances verified in Accounts Overview after fund transfer. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_002 |
| **Test Case Title** | Fund transfer with a small decimal amount ($0.02) |
| **Preconditions** | User is logged in to ParaBank (russell / test1234)<br>Source account has a positive balance |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Enter amount: $0.02<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 16123<br>Amount: $0.02 |
| **Expected Result** | Transfer processed successfully<br>Confirmation page shows $0.02<br>Both account balances updated accordingly |
| **Actual Result** | $0.02 transfer accepted and processed<br>Confirmation page displayed with correct amount<br>Balance change reflected in both accounts |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **Medium** |
| **Severity** | Major |

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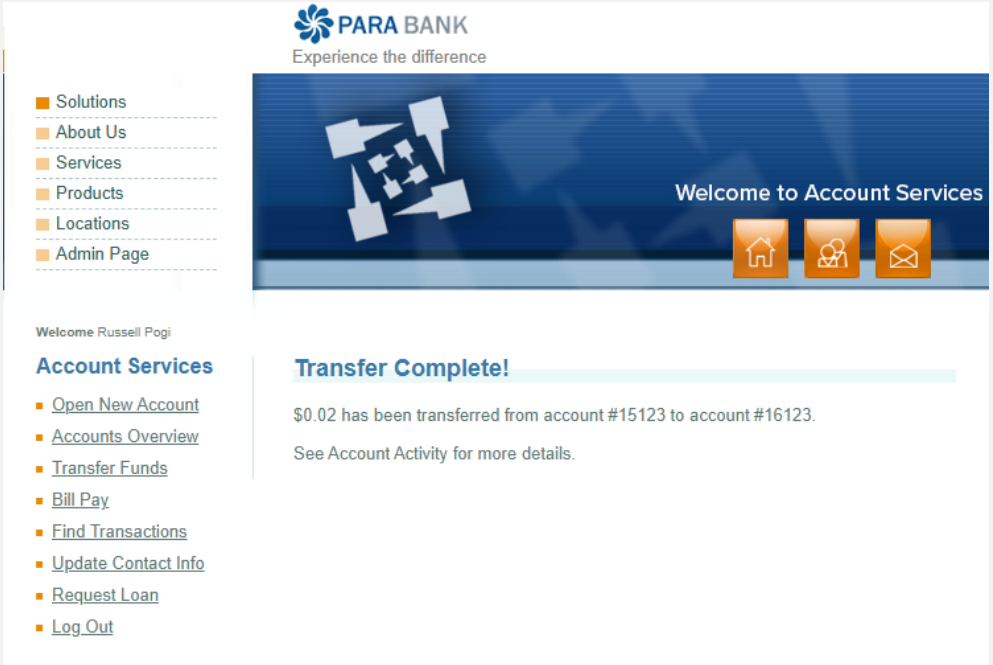
| **Remarks / Comments** |  <br>Decimal value fund transfer reflected accordingly. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_003 |
| **Test Case Title** | Fund transfer with a large valid amount |
| **Preconditions** | User is logged in<br>Source account balance is at least $500.00 |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select source account ($500.00)<br>3. Select target account<br>4. Enter amount: $499.99<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123 (Balance: atleast $500.00)<br>To Account: 16123<br>Amount: 499.99 |
| **Expected Result** | Transfer is successful<br>Confirmation page shows $499.99<br>Source balance reduced -$499.99 to its base value |
| **Actual Result** | Transfer completed successfully<br>Confirmation message displayed<br>Source account shows a reduced -$499.99 to its remaining balance |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Critical |

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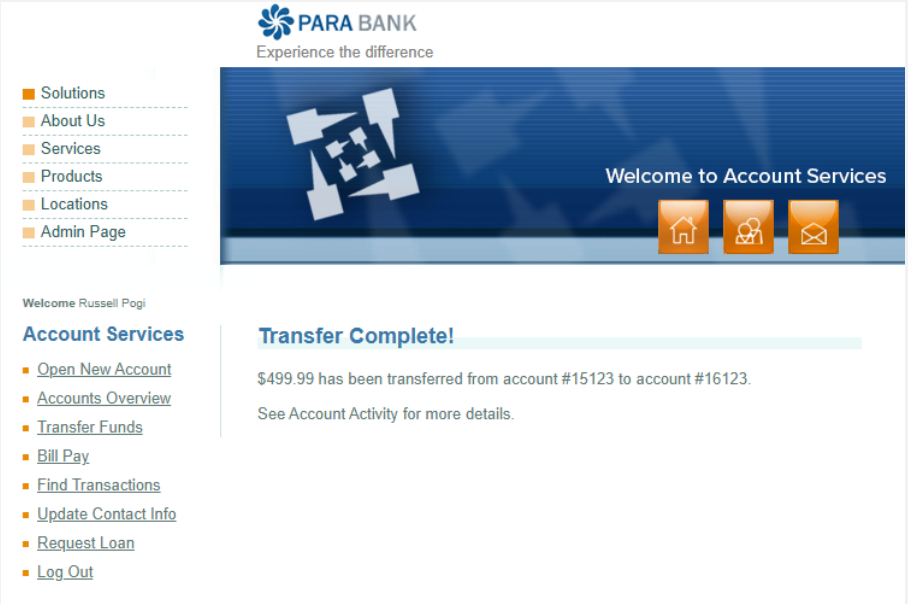
| **Remarks / Comments** |  <br>The account balance was updated accordingly |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

## TS_FT _002: Verify fund transfer with insufficient balance

| Field | Details |
|---|---|
| **Test Case ID** | TC_004 |
| **Test Case Title** | Fund transfer with amount exceeding account balance |
| **Preconditions** | User is logged in<br>Source account balance is less than $1000.00 |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select source account with less than $1000.00 balance<br>3. Select a valid target account<br>4. Enter transfer amount: $1000.00<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123 (Balance: less than $1000.00)<br>To Account: 16123<br>Amount: $1000.00 |
| **Expected Result** | System displays insufficient funds error<br>Transfer is not processed<br>Account balances remain unchanged |
| **Actual Result** | The transfer was still deemed successful despite the insufficient funds to transfer<br>Transfer was executed regardless of error<br>Both account balances changed |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |

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| **Severity** | Critical |
| **Remarks / Comments** | **Accounts Overview**<br><br>Account \| Balance* \| Available Amount<br>15123 \| -$934.55 \| $0.00<br>16123 \| $1500.05 \| $1500.05<br><br>BUG: Allowed the transfer despite insufficient funds, balance went negative. Missing server-side balance validation. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_005 |
| **Test Case Title** | Fund transfer with amount $0.01 over the available balance |
| **Preconditions** | User is logged in<br>Source account balance is known |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select source account ($1500.05)<br>3. Select valid target account<br>4. Enter amount: $1500.06<br>5. Click 'Transfer' |
| **Test Data** | From Account: 16123 (Balance: $1500.05)<br>To Account: 15123<br>Amount: $1500.06 |
| **Expected Result** | System blocks the transfer<br>Error message shown<br>Balance should remain unchanged for both |
| **Actual Result** | Transfer was processed without error despite $0.01 excess<br>Source account balance dropped to -$0.01<br>System could not block the transfer |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |
| **Severity** | Critical |

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| **Remarks / Comments** | <br>BUG: Further confirms TC_004 defect. System has no server-side validation most probably |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_006 |
| **Test Case Title** | Fund transfer from an account with $0.00 balance |
| **Preconditions** | User is logged in<br>Source account balance is $0.00 |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select source account with $0.00 balance<br>3. Select valid target account<br>4. Enter amount: $10.00<br>5. Click 'Transfer' |
| **Test Data** | From Account: 17123 (Balance: $0.00)<br>To Account: 16123<br>Amount: $10.00 |
| **Expected Result** | System should display an error of insufficient or zero balance<br>Transfer is blocked |
| **Actual Result** | Transfer processed with no error<br>Source account balance went to -$10.00<br>Consistent with defect in TC_004 and TC_005 |

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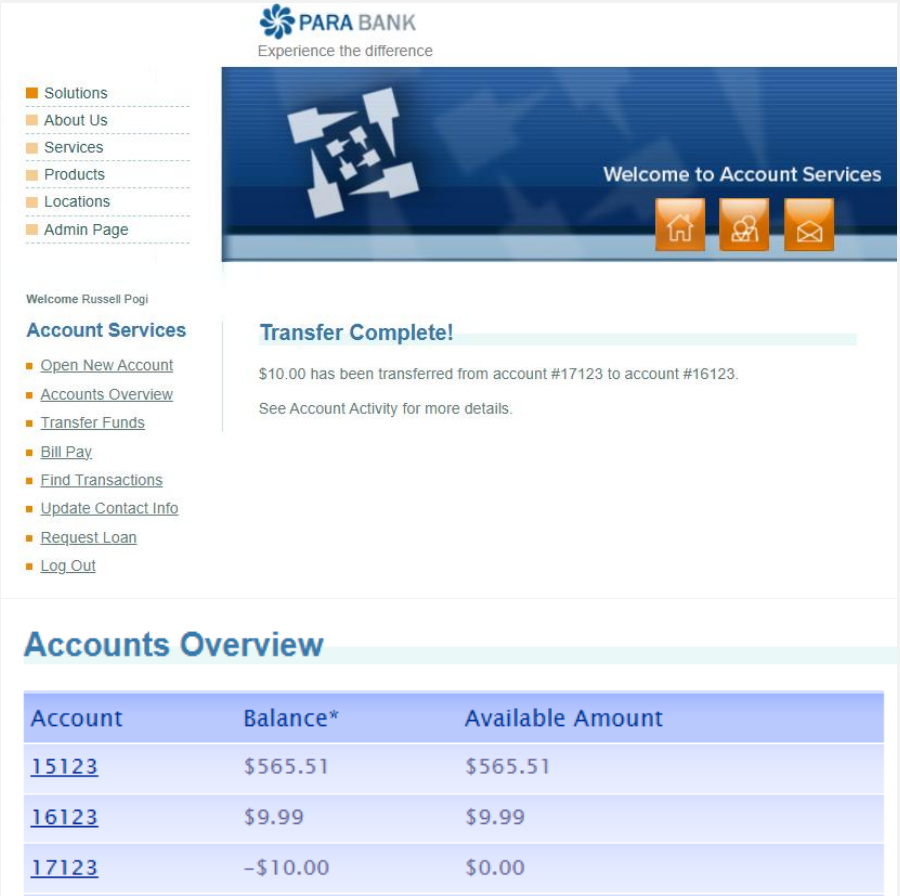
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |
| **Severity** | Critical |
| **Remarks / Comments** |  <br>BUG: Third confirmation of defective balance validation when it comes to transferring funds. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

## TS_FT _003: Verify fund transfer with invalid or missing amount

| Field | Details |
|---|---|
| **Test Case ID** | TC_007 |
| **Test Case Title** | Fund transfer with empty amount field |
| **Preconditions** | User is logged in<br>User has at least two active accounts |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Leave Amount field blank<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123 |

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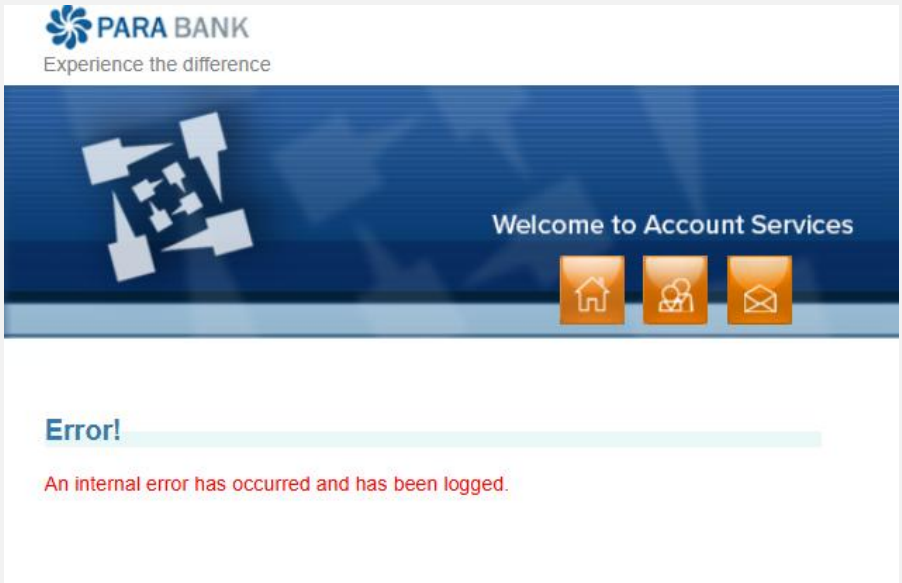
| | To Account: 16123<br>Amount: (empty) |
| **Expected Result** | Required field validation error shown on Amount field<br>Transfer is not pushed through |
| **Actual Result** | System displayed that an error has occurred<br>Transfer was not submitted<br>However, there was no mention of which field caused the error |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Major |
| **Remarks / Comments** |  <br>There is somewhat a validation for empty transfer amount field, however it is not specified on the error message, the user might think that the whole platform is down or something like that. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_008 |
| **Test Case Title** | Fund transfer with zero amount |
| **Preconditions** | User is logged in<br>User has at least two active accounts |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Enter amount: 0<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 16123 |

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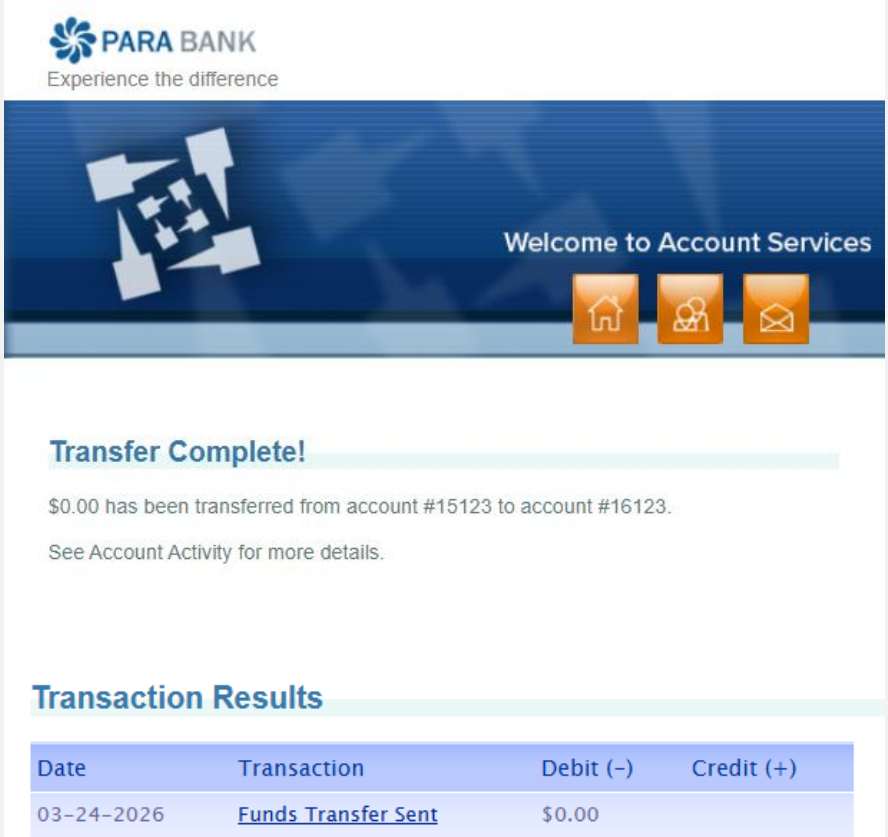
| | Amount: 0 |
| **Expected Result** | System should show a validation error that the amount must be greater than zero<br>Transfer not processed |
| **Actual Result** | System accepted $0 and displayed 'Transfer Complete!'<br>No validation error shown<br>A $0.00 transaction record was created in account activity |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |
| **Severity** | Major |
| **Remarks / Comments** |  <br>BUG: Zero-amount transfers are not blocked, and transaction records are created. It should not be possible and must be handled appropriately. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_009 |
| **Test Case Title** | Fund transfer with a negative amount |
| **Preconditions** | User is logged in<br>User has at least two active accounts |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account |

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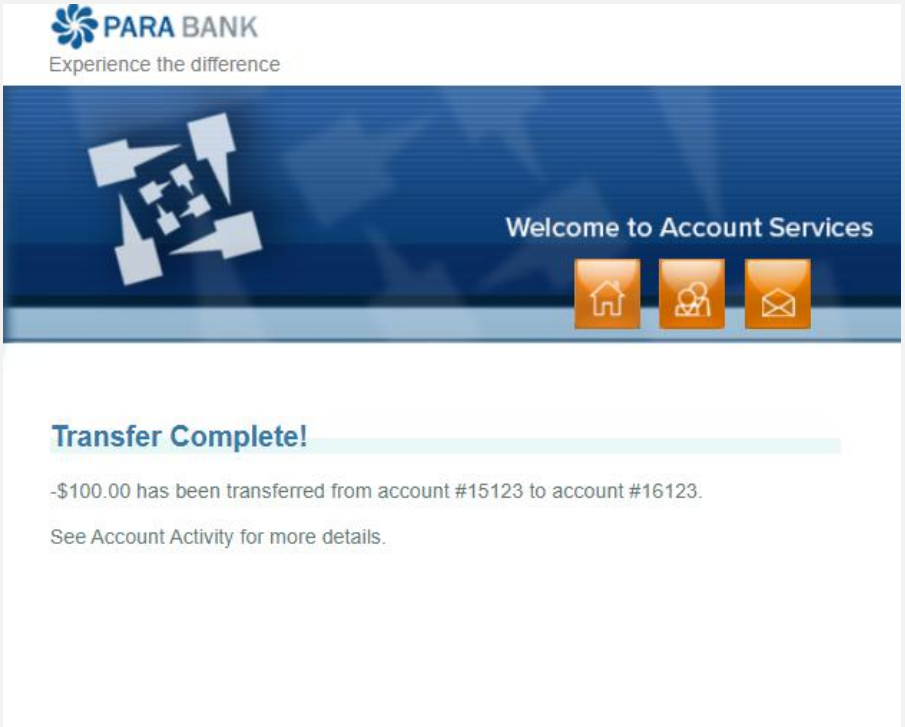
|---|---|
| | 3. Select valid To Account<br>4. Enter amount: -100<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 16123<br>Amount: -100 |
| **Expected Result** | System rejects negative value<br>Validation error displayed<br>Transfer not executed |
| **Actual Result** | The system accepted a negative value<br>There was no validation for the transaction went through<br>Transfer was executed |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |
| **Severity** | Critical |
| **Remarks / Comments** |  <br>BUG: There is an absence of validation for negative fund amount, it should not be possible on a fund transfer from user to user. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_010 |
| **Test Case Title** | Fund transfer with non-numeric characters in amount |
| **Preconditions** | User is logged in<br>User has at least two active accounts |

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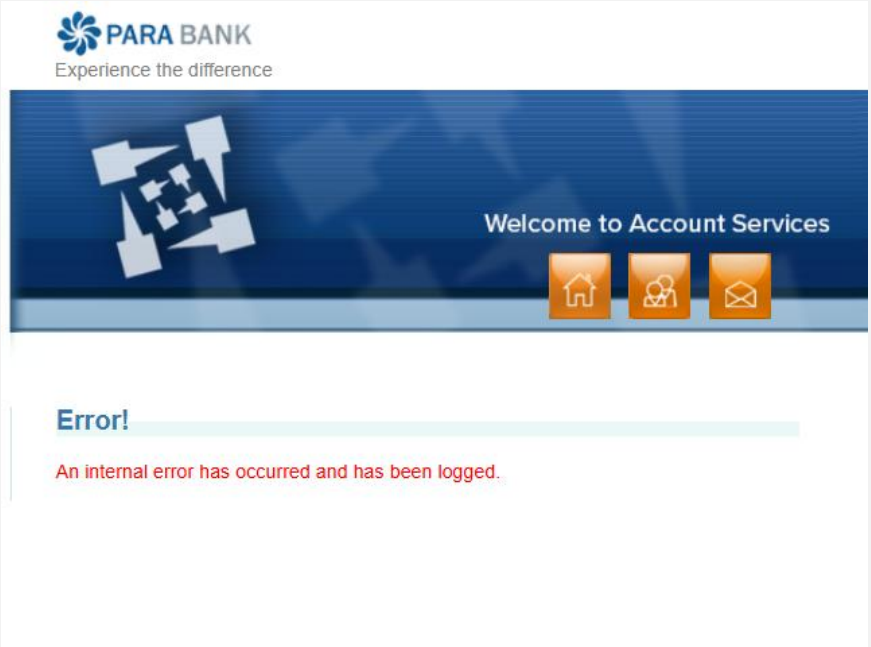
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Enter amount: abc<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 16123<br>Amount: abc |
| **Expected Result** | System should show a validation error that amount must be numeric<br>Transfer not processed |
| **Actual Result** | System displayed that an error has occurred<br>Transfer was not submitted<br>However, there was no mention of what action caused the error, like TC_007 |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **Medium** |
| **Severity** | Major |
| **Remarks / Comments** |  <br>Similar to TC_007 there is somewhat a validation for non-numeric characters in transfer amount field, however it is not specified on the error message, the user will never know what went wrong/caused the error. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_011 |
| **Test Case Title** | Fund transfer with special characters in amount |
| **Preconditions** | User is logged in |

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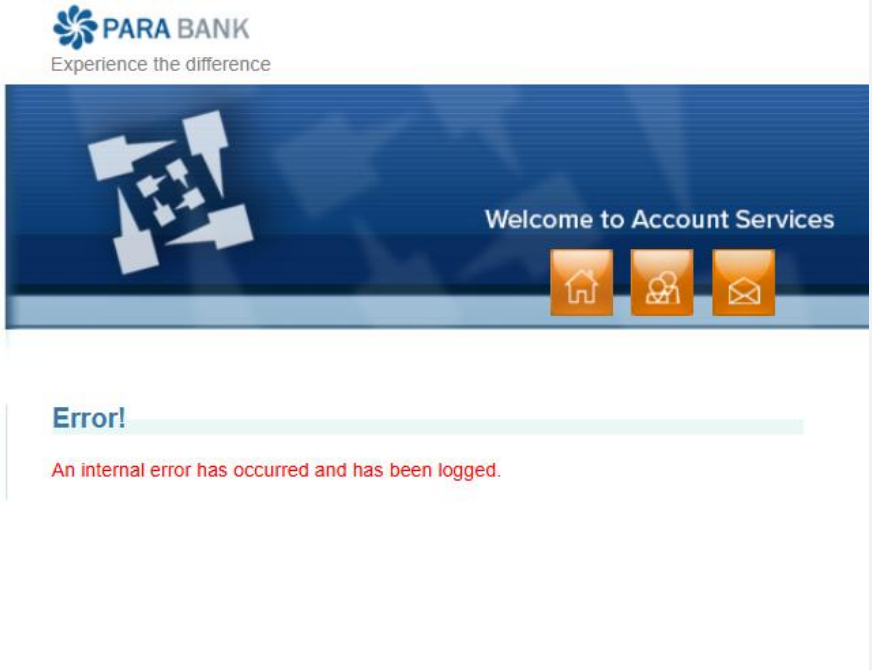
|---|---|
| | User has at least two active accounts |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Enter amount: !@#<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 16123<br>Amount: !@# |
| **Expected Result** | Special characters are rejected or stripped<br>No transfer is submitted |
| **Actual Result** | System displayed that an error has occurred<br>Transfer was not submitted<br>However, there was no mention of what action caused the error, like TC_007, TC_010 |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **Low** |
| **Severity** | Minor |
| **Remarks / Comments** |  <br>Same case with TC_007 and TC_010 there is somewhat a validation for special characters in transfer amount field, however it is not specified on the error message, the user will never know what went wrong/caused the error. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

## TS_FT_004: Verify fund transfer to the same account

| Field | Details |
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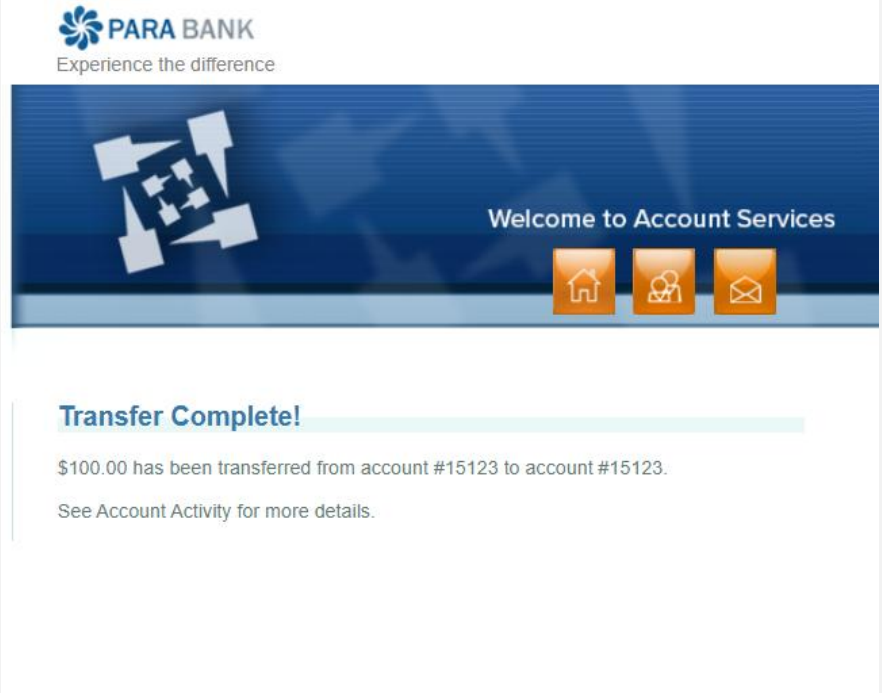
| **Test Case ID** | TC_012 |
| **Test Case Title** | Fund transfer where From and To account is identical |
| **Preconditions** | User is logged in<br>User has at least one active account |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select account 15123 as From Account<br>3. Select same account 15123 as To Account<br>4. Enter amount: $100.00<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 15123 (same)<br>Amount: $100.00 |
| **Expected Result** | System prevents same-account transfer<br>Error message displayed<br>Transfer not processed |
| **Actual Result** | The To Account dropdown did not exclude the source account from its options<br>User was able to select the same account in both fields<br>Transfer was processed, balance unchanged but a transaction record was created |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **Medium** |
| **Severity** | Major |
| **Remarks / Comments** |  <br>BUG: Same-account transfers are not blocked. Client-side should filter the source account out of the destination dropdown for a straightforward approach to this validation. |

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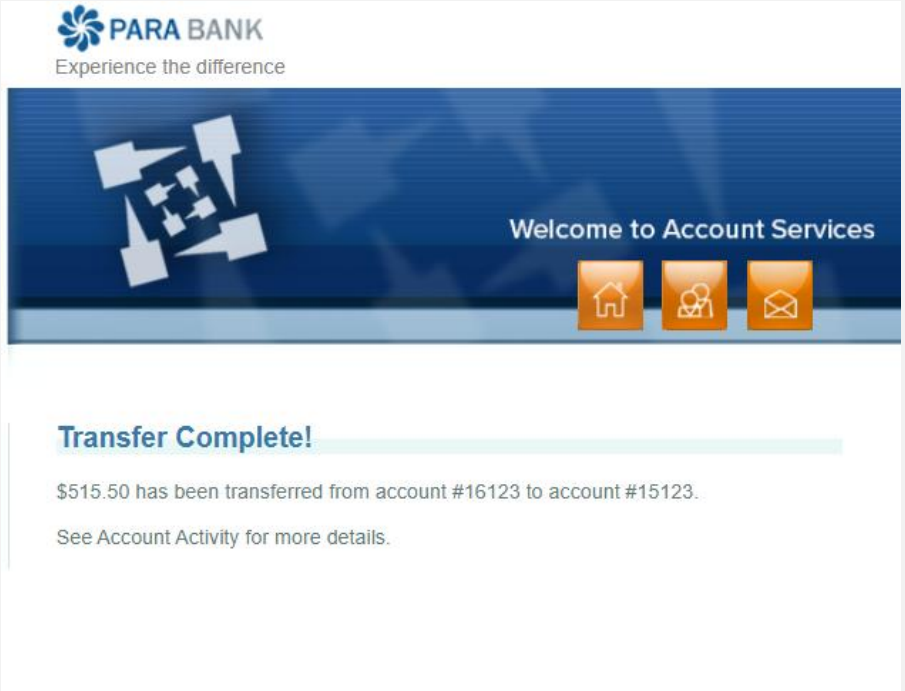
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_013 |
| **Test Case Title** | Verify To Account dropdown excludes the selected From Account |
| **Preconditions** | User is logged in<br>User has multiple accounts |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select account 15123 from 'From Account'<br>3. Inspect available options in 'To Account' dropdown |
| **Test Data** | From Account: 15123<br>To Account: observe dropdown options |
| **Expected Result** | Account 15123 should not appear in the To Account dropdown after being selected as source |
| **Actual Result** | Account 15123 was still visible and selectable in the To Account dropdown<br>No dynamic filtering was applied when From Account selection changed |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **Medium** |
| **Severity** | Major |
| **Remarks / Comments** | <br>BUG: A defect in terms of the design philosophy, similar to my remarks on TC_012, one should not be able to transfer funds to oneself, and the client side should handle it by removing it from the dropdown. |
| **Tested By / Date** | Russell Bernardo / 2026-03-24 |

## TS_FT_005: Verify fund transfer with boundary amount values

| Field | Details |
|---|---|
| **Test Case ID** | TC_014 |
| **Test Case Title** | Fund transfer with exact available balance |
| **Preconditions** | User is logged in<br>Source account balance is exactly $515.50 |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select source account ($515.50)<br>3. Select valid target account<br>4. Enter amount: $515.50<br>5. Click 'Transfer' |
| **Test Data** | From Account: 16123 (Balance: $515.50)<br>To Account: 15123<br>Amount: $415.50 |
| **Expected Result** | Transfer is successful<br>Source balance becomes $0.00<br>Confirmation page displayed |
| **Actual Result** | Transfer completed successfully for the full $515.50<br>Confirmation page displayed with correct amount<br>Source account balance now shows $0.00 in Accounts Overview |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Critical |
| **Remarks / Comments** |  |

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| | <br>Accounts Overview<br>Account \| Balance* \| Available Amount<br>15123 \| $515.50 \| $515.50<br>16123 \| $0.00 \| $0.00<br>Total $515.50<br>*Balance includes deposits that may be subject to holds<br>Full balance transfers are allowed. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_015 |
| **Test Case Title** | Fund transfer with minimum valid amount ($0.01) |
| **Preconditions** | User is logged in<br>Source account has a positive balance |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Enter amount: $0.01<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123<br>To Account: 16123<br>Amount: $0.01 |
| **Expected Result** | $0.01 transfer accepted and processed<br>Confirmation page displayed<br>Transaction logged in account activity |
| **Actual Result** | $0.01 transfer was accepted and processed<br>Confirmation page showed correct amount<br>Transaction logged in account activity |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **Medium** |
| **Severity** | Minor |

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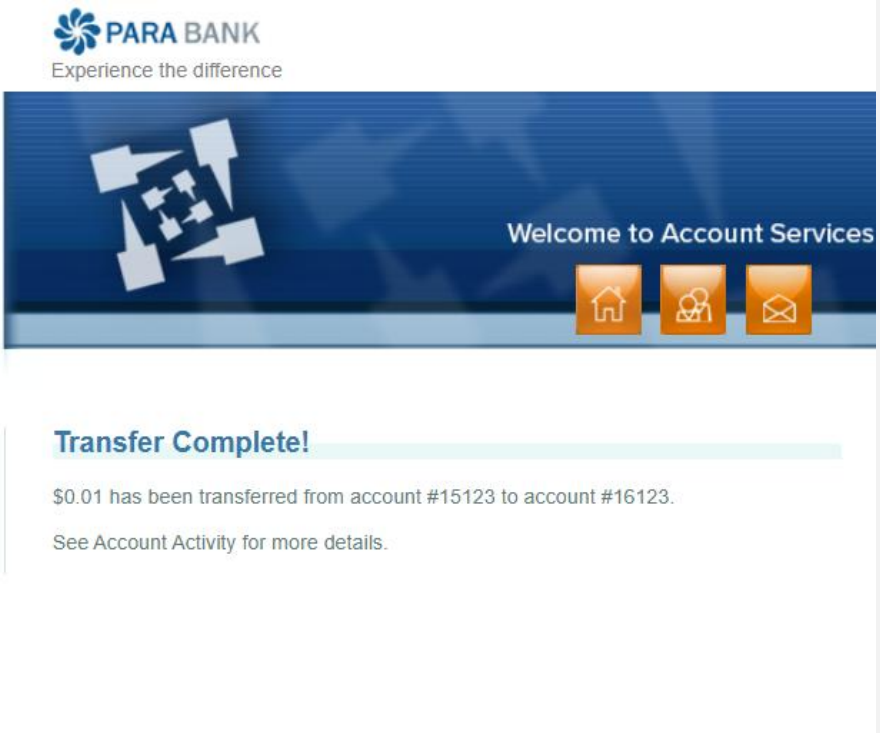
| **Remarks / Comments** |  <br>Minimum valid amount processes correctly. No lower limit enforced below this value, the zero-amount issue is covered in TC_008. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_016 |
| **Test Case Title** | Fund transfer with amount $0.01 over available balance |
| **Preconditions** | User is logged in<br>Source account balance is exactly $515.49 |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select source account ($515.49)<br>3. Select valid target account<br>4. Enter amount: $515.50<br>5. Click 'Transfer' |
| **Test Data** | From Account: 15123 (Balance: $515.49)<br>To Account: 16123<br>Amount: $515.50 |
| **Expected Result** | System blocks the transfer — amount exceeds available balance<br>Error message displayed<br>Balance unchanged |
| **Actual Result** | Transfer was processed without error<br>Balance dropped to -$0.01<br>Consistent with overdraft defect in TC_004 and TC_005 |

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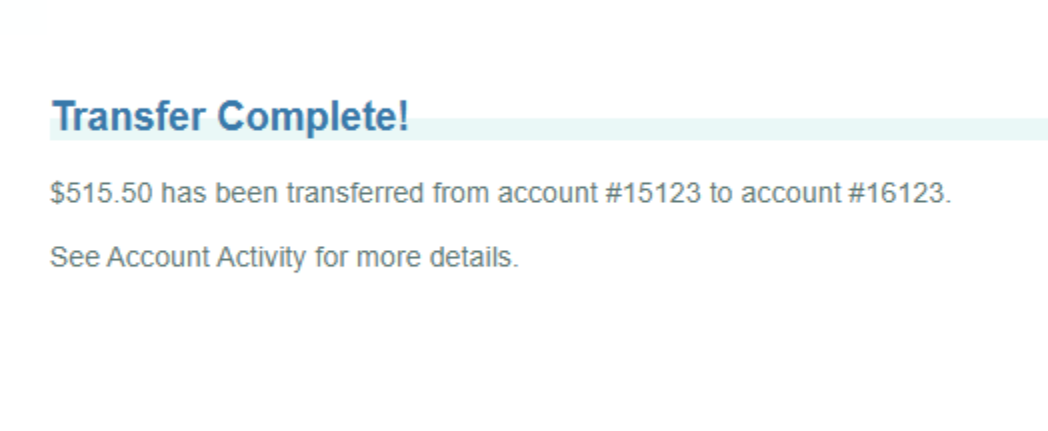
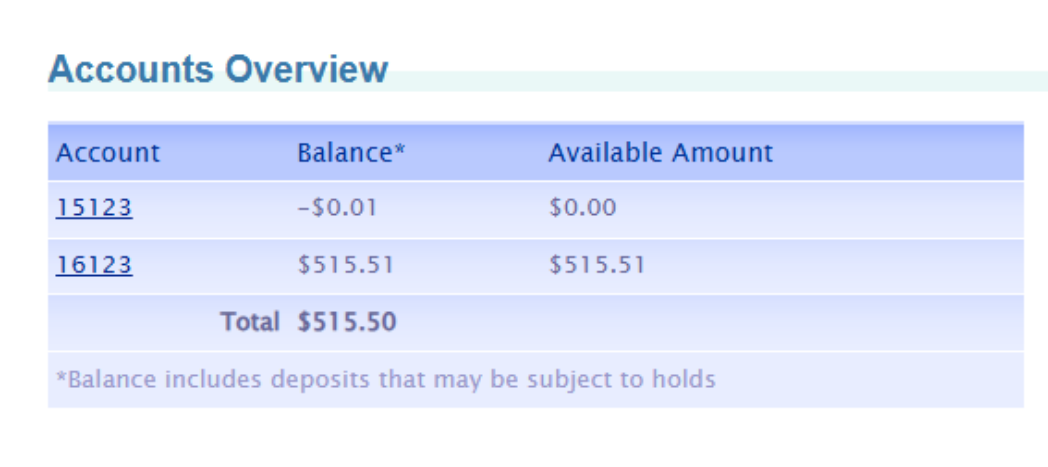
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |
| **Severity** | Critical |
| **Remarks / Comments** |  <br>  <br>Accounts Overview<br>Account \| Balance* \| Available Amount<br>15123 \| -$0.01 \| $0.00<br>16123 \| $515.51 \| $515.51<br>Total $515.50<br>*Balance includes deposits that may be subject to holds<br>BUG: Boundary value confirms vulnerability. System allows transfers $0.01 over balance. Linked to defects in TC_004 and TC_005. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_017 |
| **Test Case Title** | Fund transfer with a very large amount (stress value) |
| **Preconditions** | User is logged in |
| **Test Steps** | 1. Navigate to Transfer Funds<br>2. Select valid From Account<br>3. Select valid To Account<br>4. Enter amount: 999999999999999<br>5. Click 'Transfer' |
| **Test Data** | From Account: 16123<br>To Account: 15123 |

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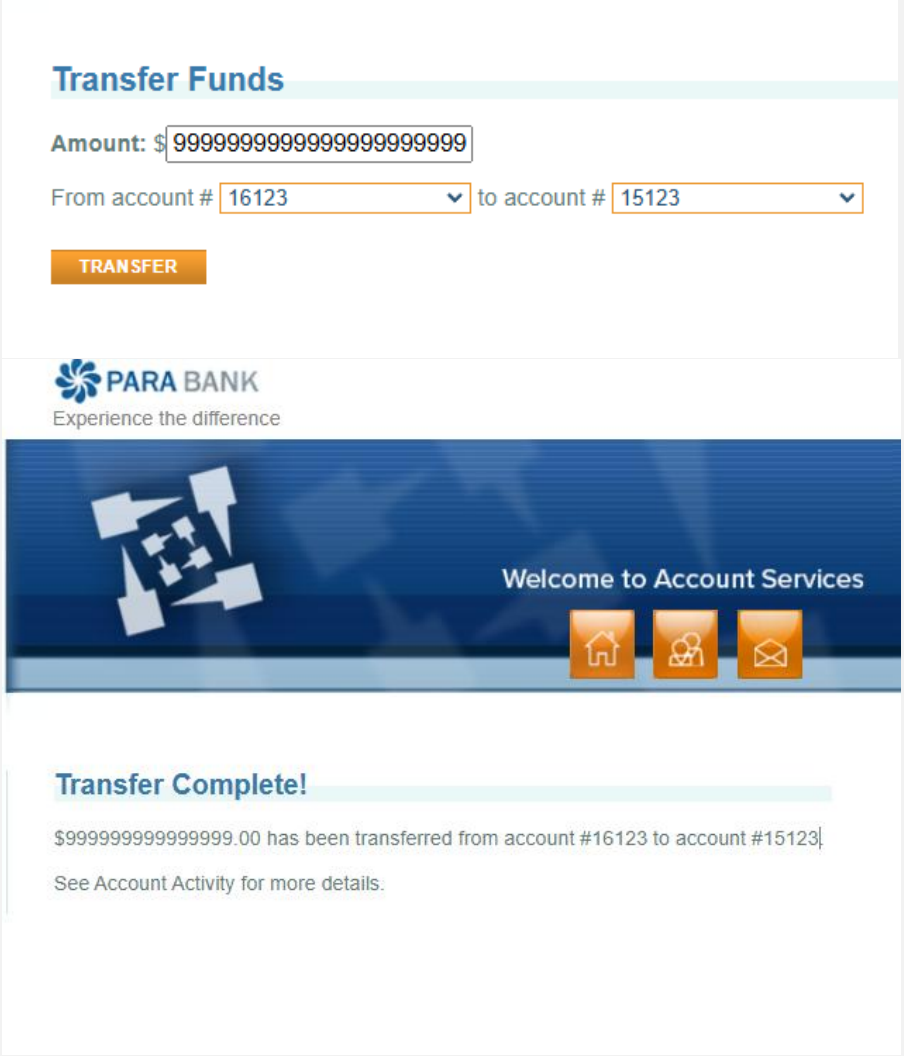
| | Amount: 999999999999999 |
| **Expected Result** | System enforces a maximum transfer limit or blocks due to insufficient balance<br>No system crash or unhandled error |
| **Actual Result** | System accepted the value and processed the transfer<br>No maximum transfer limit was enforced<br>Application remained stable, no crash, but as expected it created negative transaction |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **Medium** |
| **Severity** | Major |
| **Remarks / Comments** |  |

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| | <br>Accounts Overview<br>Account \| Balance* \| Available Amount<br>15123 \| $999999999999999.00 \| $999999999999999.00<br>16123 \| -$999999999999483.50 \| $0.00<br>Total $515.50<br>*Balance includes deposits that may be subject to holds<br><br>BUG: No upper limit on transfer amount. Combined with other validation defects, this could allow arbitrarily large negative balances. It should have a maximum/minimum transfer limit as to how modern digital banks implement it. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

## TS_FT_006: Verify transfer confirmation and transaction record

| Field | Details |
|---|---|
| **Test Case ID** | TC_018 |
| **Test Case Title** | Verify confirmation page content after successful transfer |
| **Preconditions** | User is logged in<br>A valid transfer of $100.00 has been submitted |
| **Test Steps** | 1. Complete a valid transfer of $100.00 from account 17123 to 18123<br>2. Observe the confirmation page content |
| **Test Data** | From Account: 17123<br>To Account: 18123<br>Amount: $100.00 |
| **Expected Result** | Page shows 'Transfer Complete!'<br>Amount, source, and destination accounts are displayed<br>A transaction ID link is present |
| **Actual Result** | Confirmation page displayed 'Transfer Complete!'<br>Amount $100.00 shown correctly<br>Both account numbers listed<br>A clickable transaction ID link was present |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Major |

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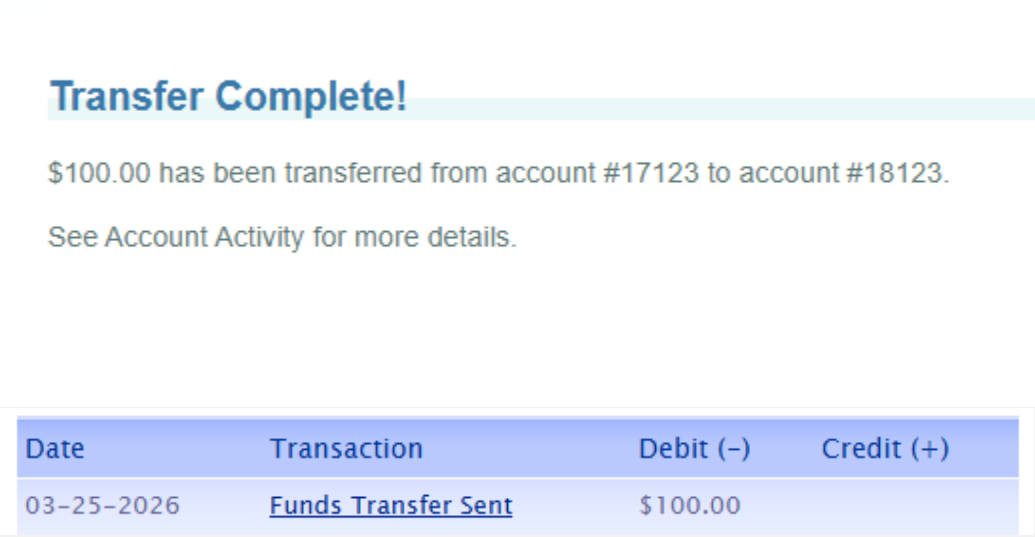
|---|---|
| **Remarks / Comments** |  <br>All expected confirmation elements are present. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_019 |
| **Test Case Title** | Verify debit transaction appears in source account activity |
| **Preconditions** | A successful fund transfer of $100.00 from account 17123 has been completed |
| **Test Steps** | 1. Navigate to Accounts Overview<br>2. Click on account 17123<br>3. Review the Account Activity section |
| **Test Data** | Account: 17123<br>Expected transaction type: Debit<br>Expected amount: -$100.00 |
| **Expected Result** | Debit transaction for $100.00 visible in account activity<br>Labeled as 'Funds Transfer Sent'<br>Transaction date matches today's date |
| **Actual Result** | Debit transaction for $100.00 visible in account 17123 activity<br>Labeled as 'Funds Transfer Sent'<br>Date and amount correct |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Major |

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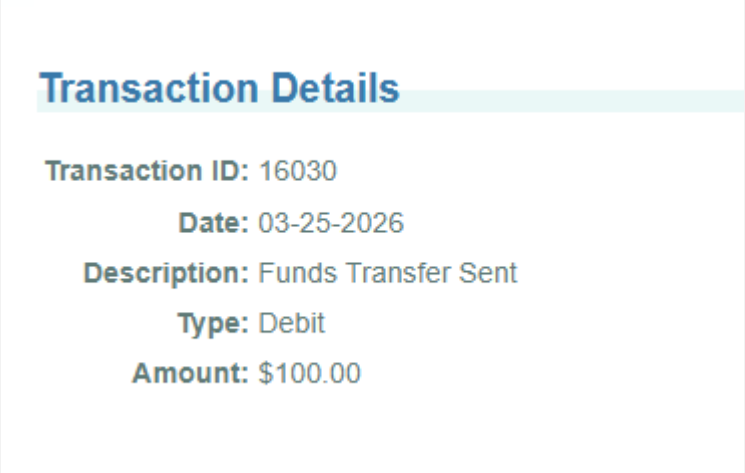
| **Remarks / Comments** |  <br>Transaction amount is accurate and immediately visible after the transfer with no page refresh required. However, both debit(sender) and credit(receiver) transactions are defaulted as debit and as a ‘Fund transfer sent’, this would be a bug for the receiver side. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |



| Field | Details |
|---|---|
| **Test Case ID** | TC_020 |
| **Test Case Title** | Verify credit transaction appears in destination account activity |
| **Preconditions** | A successful fund transfer of $100.00 to account 18123 has been completed |
| **Test Steps** | 1. Navigate to Accounts Overview<br>2. Click on account 18123<br>3. Review the Account Activity section |
| **Test Data** | Account: 18123<br>Expected transaction type: Credit<br>Expected amount: +$100.00 |
| **Expected Result** | Credit transaction for $100.00 visible in account activity<br>Labeled as 'Funds Transfer Received'<br>Transaction date matches today's date |
| **Actual Result** | The transaction for $100.00 visible in account 18123 activity is still labeled as debit instead of credit<br>Labeled as 'Funds Transfer Sent’ instead of received<br>Date and amount correct |
| **Status (Pass/Fail)** | **Fail** |
| **Priority** | **High** |
| **Severity** | Major |

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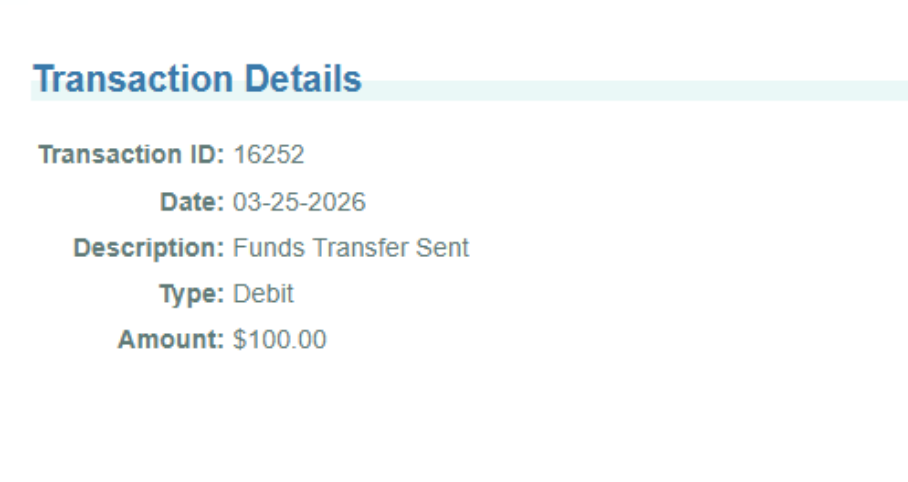
| **Remarks / Comments** |  <br>Transaction Details<br>Transaction ID: 16252<br>Date: 03-25-2026<br>Description: Funds Transfer Sent<br>Type: Debit<br>Amount: $100.00<br>BUG: As mentioned in TC_019 the receiver of fund transfer in terms of labeling and such is not properly accorded. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_021 |
| **Test Case Title** | Verify account balance updates immediately after transfer |
| **Preconditions** | User is logged in<br>Source account balance is $415.50 before transfer |
| **Test Steps** | 1. Note source account balance: $415.50<br>2. Complete a transfer of $100.00<br>3. Navigate to Accounts Overview<br>4. Observe the displayed balance for the source account |
| **Test Data** | From Account: 17123 (Before: $415.50)<br>To Account: 18123<br>Amount: $100.00 |
| **Expected Result** | Source account balance updates to $315.50 immediately<br>No page refresh required |
| **Actual Result** | Balance updated to $315.50 on Accounts Overview without requiring a refresh<br>Change reflected within the same session immediately after transfer completion |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **Medium** |
| **Severity** | Minor |

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| **Remarks / Comments** | **Accounts Overview**<br><br>| Account | Balance* | Available Amount |<br>|---|---|---|<br>| 17123 | $315.50 | $315.50 |<br>| 18123 | $200.00 | $200.00 |<br>| Total | $515.50 | |<br><br>*Balance includes deposits that may be subject to holds<br><br>Real-time balance update confirmed. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

## TS_FT_007: Verify access control for unauthenticated users

| Field | Details |
|---|---|
| **Test Case ID** | TC_022 |
| **Test Case Title** | Unauthenticated user direct navigation to Transfer Funds page |
| **Preconditions** | User is NOT logged in<br>Browser has no ParaBank session cookies (use Incognito) |
| **Test Steps** | 1. Open a new Incognito browser window<br>2. Navigate directly to: http://parabank.parasoft.com/parabank/transfer.htm<br>3. Observe system response |
| **Test Data** | URL: /parabank/transfer.htm<br>Session: None (unauthenticated) |
| **Expected Result** | User is redirected to login page<br>Transfer Funds page is not displayed<br>No account data is exposed |
| **Actual Result** | Browser was immediately redirected to the login page<br>Transfer Funds page content was not shown |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Critical |

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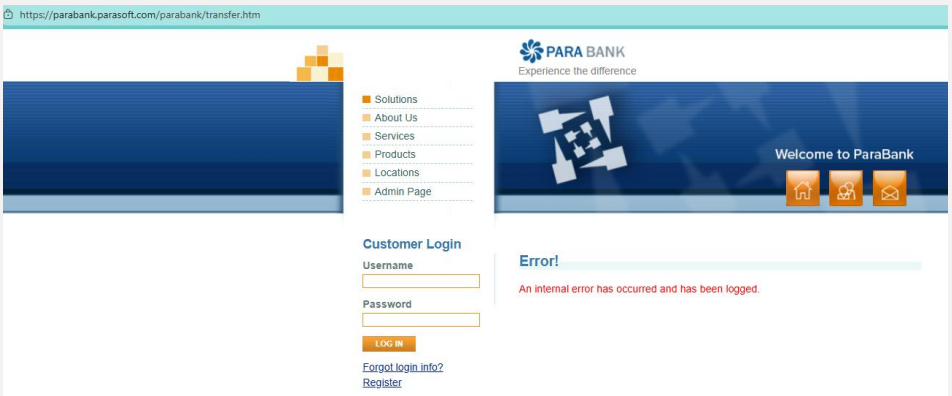
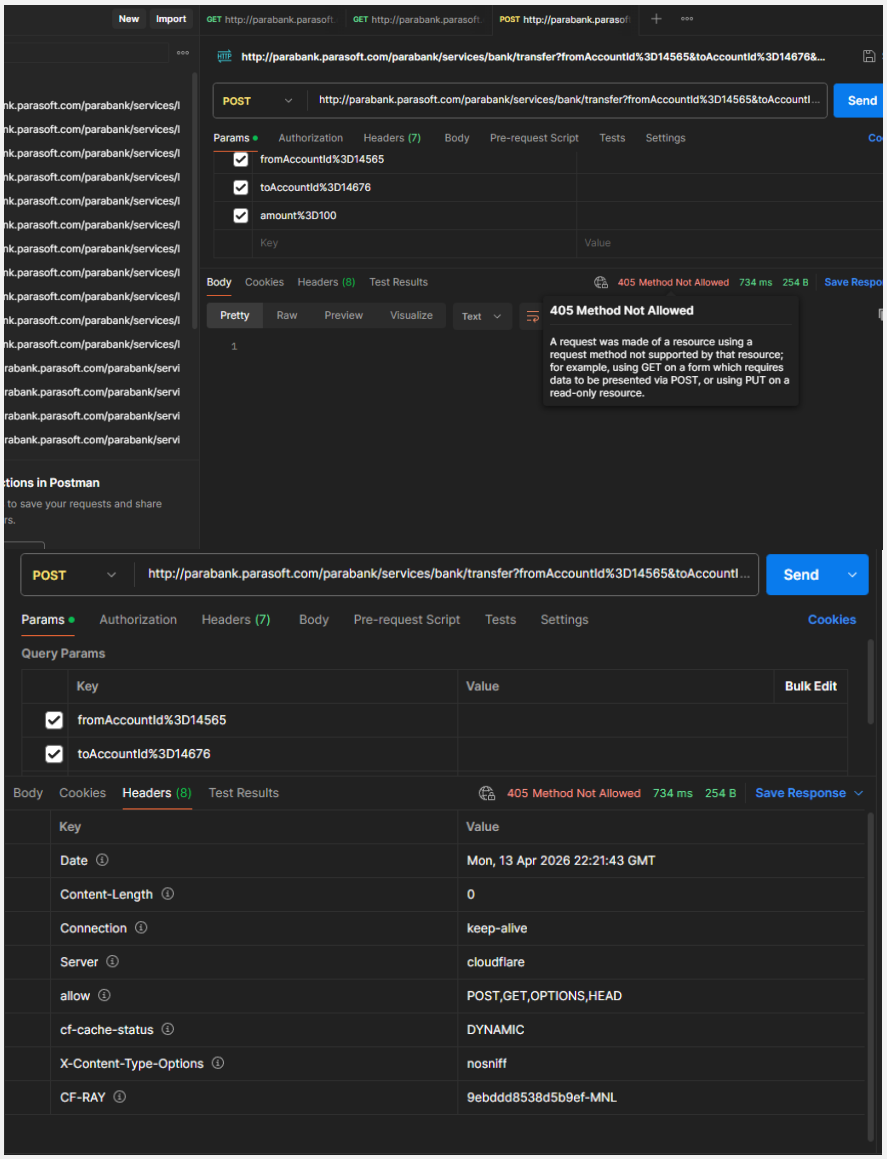
|---|---|
| **Remarks / Comments** |  <br>Access control is enforced for UI navigation. Direct URL access is correctly blocked for unauthenticated sessions. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_023 |
| **Test Case Title** | Unauthenticated API call to POST /transfer endpoint |
| **Preconditions** | Postman is open<br>No authentication cookies or headers are set |
| **Test Steps** | 1. Create a new POST request in Postman<br>2. Set URL: /parabank/services/bank/transfer<br>3. Add valid params — ensure no auth cookies or headers are included<br>4. Click Send |
| **Test Data** | Method: POST<br>fromAccountId: 14565<br>toAccountId: 14676<br>amount: 100<br>Auth: None |
| **Expected Result** | HTTP 401 Unauthorized<br>Transfer is not executed<br>No account data returned |
| **Actual Result** | HTTP 405 Method Not Allowed<br>Transfer is not executed<br>No account data is manipulated |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Critical |

| | |
|---|---|
| **Remarks / Comments** |  <br><br>The web version of ParaBank is protected by cloudflare which is why it is difficult to penetrate it using endpoints for the session is always stripped, per the CF-RAY header. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |

| Field | Details |
|---|---|
| **Test Case ID** | TC_024 |
| **Test Case Title** | Attempt transfer after logout |
| **Preconditions** | User was previously logged in and has now logged out |
| **Test Steps** | 1. Log in to ParaBank<br>2. Navigate to Transfer Funds<br>3. Log out<br>4. Press browser Back button to return to the Transfer Funds page<br>5. Attempt to submit a transfer |

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| **Test Data** | Username: russell<br>Action: back navigation after logout |
| **Expected Result** | Cached page is not functional<br>Transfer attempt redirects to login page<br>No transfer executed with expired session |
| **Actual Result** | Browser back button showed the cached Transfer Funds page<br>Filling the form and clicking Transfer redirected to the login page<br>No transfer was processed |
| **Status (Pass/Fail)** | **Pass** |
| **Priority** | **High** |
| **Severity** | Critical |
| **Remarks / Comments** | <br>Session invalidation works correctly. Even if the cached UI page is accessible, the server rejects requests with an expired session and redirects to login page. |
| **Tested By / Date** | Russell Bernardo / 2026-03-25 |